

Are Inflation Expectations Better Anchored Under Fixed Exchange Rate Regimes? Evidence from Denmark and Sweden

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Abstract

This paper asks whether inflation expectations are better anchored under fixed exchange rate regimes. The theoretical motivation is to test whether a fixed exchange rate can serve as an external nominal anchor by tying domestic monetary policy to a more credible monetary authority. I test this idea by comparing Denmark's fixed with Sweden's floating exchange rate regime. Using quarterly data from 1995Q4 to 2025Q2, I estimate an instrumental variable local projection model to measure the dynamic response of inflation expectations to inflation shocks, with realized inflation instrumented using oil supply news shocks. The results do not provide evidence that inflation expectations are more strongly anchored under Denmark's fixed exchange rate regime than under Sweden's floating regime. The findings suggest that fixed exchange rates alone do not guarantee stronger anchoring in inflation expectations, especially when shocks are global and when floating-regime countries have credible monetary institutions.

Keywords: Inflation Expectations, Fixed Exchange Rates, Monetary Credibility, Local Projections, Denmark, Sweden

JEL Codes: E31, E52, F31, F33

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1 Introduction

Inflation expectations play a vital role in today’s monetary policy because they affect how households, firms, and markets respond to inflation. When economic agents believe inflation will return to a stable long-run level, temporary inflation shocks should have a very limited effect on long-term expectations of future inflation, which is what it means for expectations to be “anchored”. This means that inflation expectations can act as a stabilizing force as they reduce the risk of short-run inflation becoming persistent in household beliefs. When expectations are weakly anchored, realized inflation shocks may lead agents to revise beliefs more strongly, making inflation more persistent and harder for policymakers to control.

This paper asks whether inflation expectations are better anchored under fixed exchange rate regimes. A fixed exchange rate can act as an external nominal anchor by tying domestic monetary policy to a larger and potentially more credible monetary authority. Theoretically, a country’s commitment to pegging its currency can strengthen policy credibility. If households believe the exchange rate peg is credible, an important question is whether they would be less likely to revise their inflation expectations in response to temporary inflation shocks. In contrast, for a floating exchange rate regime, inflation expectations could depend more directly on the credibility of the domestic central bank and its inflation targeting framework.

I examined the role of monetary regimes in anchoring inflation expectations by comparing Denmark and Sweden. The two countries provide a useful comparison because they are both well-developed Scandinavian economies with similar institutions, data availability, and exposure to global shocks, but they have different monetary regimes. Denmark adopted its current fixed exchange rate regime in 1982 by pegging to the Deutsche Mark, and to the Euro in 1999. Sweden uses a floating exchange rate regime with an independent inflation-targeting central bank. The difference in monetary regimes allows the paper to examine whether inflation expectations respond differently to inflation shocks under different

monetary frameworks.

Empirically, I estimate the dynamic response of inflation expectations based on inflation shocks using an instrumental variable local projection framework. I use inflation expectations from the European Commission consumer survey, and I use CPI to measure inflation. To address the endogeneity of inflation, I use [Känzig \(2021\)](#) oil supply news shock as an external instrument for realized inflation. I estimate impulse responses separately for Denmark and Sweden and then test whether the response paths differ across countries.

My results do not provide evidence that inflation expectations are more strongly anchored under Denmark’s fixed exchange rate regime than under Sweden’s floating regime. Sweden initially exhibits a larger reaction to inflation shocks, Denmark shows more persistence at medium and longer horizons. However, after a joint Wald test, we fail to reject the null hypothesis that the Danish and Swedish inflation expectation response paths are equal, given that the overall impulse response paths are not statistically different.

These findings suggest that the presence of a fixed exchange rate alone does not guarantee more strongly anchored inflation expectations. One interpretation is that the Danish peg to the euro does not fully insulate expectations from global inflation shocks, especially given those shocks also impact the Euro as well. Another perspective is that Swedish monetary institutions are sufficiently credible that its floating exchange rate regime may anchor expectations as well as Denmark’s fixed exchange rate regime. Generally speaking, the results suggest that monetary credibility may depend less on the exchange rate regime itself, but more on the institutional environment supporting the regime.

The rest of the paper is as follows. Section 2 is a literature review on inflation expectations, monetary credibility, central bank communication, and credibility from fixed exchange rates. Section 3 describes the data collection process and how I processed the data. Section 4 outlines the instrumental variable local projection framework. Section 5 presents the impulse response results, significance tests, and robustness checks. Section 6 concludes by interpret-

ing the findings, discussing their implications for the current literature, and examining their policy implications.

2 Literature Review

This literature review places this paper in the topics of inflation expectations, monetary credibility, and exchange-rate regimes. First, I outline the theoretical role of inflation expectations in standard macroeconomic models, where forward-looking expectations are related to inflation dynamics through the expectations-augmented Phillips curve and the New Keynesian Phillips curve. This theoretical framework motivates my paper’s empirical interpretation of anchoring: if expectations are well anchored, they should respond weakly to inflation shocks; if they are less anchored, they should respond more strongly. The literature review then turns to recent evidence showing that inflation expectations are often imperfectly anchored, especially when households respond to salient inflation experiences, forecast errors, or periods of high inflation. Next, I discuss how anchoring depends on credibility and stability, before discussing the literature on central bank communication, household attention, and trust. I then examine past literature on fixed exchange rates as external nominal anchors, which looks at whether pegged regimes may import credibility from a more stable monetary authority. Finally, I discuss the literature that motivates my empirical approach.

The theoretical motivation for this paper begins with inflation expectations in modern macroeconomic models. In the expectations-augmented Phillips curve, actual inflation depends on expected future inflation, which makes expectations an important channel through which macroeconomic shocks can influence inflation dynamics:

$$\pi_t = \pi_t^e - \alpha(u_t - u_t^*) + \varepsilon_t, \tag{1}$$

where π_t is actual inflation, π_t^e is expected inflation, $u_t - u_t^*$ is the unemployment gap, and ε_t is a supply shock.

Similarly, the New Keynesian Phillips Curve shows that firms' current pricing decisions are a function of expectations of future inflation and marginal costs:

$$\pi_t = \beta E_t \pi_{t+1} + \kappa x_t + u_t, \quad (2)$$

where $E_t \pi_{t+1}$ is expected future inflation, x_t is the output gap or real marginal cost, and κ is the slope of the Phillips curve (Clarida et al., 1999; Woodford, 2003). This makes the stability of inflation expectations vital to macroeconomic outcomes. When expectations are well anchored, temporary inflation shocks would have very limited effects on beliefs about future inflation, while weakly anchored expectations should respond more strongly to incoming inflation data (Bernanke, 2007; Carvalho et al., 2023). Thus, the response of inflation expectations to inflation shocks is a good empirical representation of anchoring, since lower sensitivity suggests that agents continue to believe in the stability of the long-run monetary regime (Carvalho et al., 2023). This paper builds on this idea by examining whether this inflation expectation sensitivity is different across monetary regimes. I do so by comparing a fixed exchange rate regime to a floating, inflation-targeting regime.

Having established why inflation expectations matter for inflation dynamics, the next question is what it means for those expectations to be anchored. Bernanke (2007) describes anchored expectations as expectations that are relatively insensitive to incoming data, meaning that inflation surprises should not strongly affect long-run inflation beliefs. Carvalho et al. (2023) define anchoring as beliefs consistent with the policy regime, and unanchoring to have occurred when agents doubt that inflation will return to the stable long-run inflation mean. The definition this paper will use is that expectations are anchored when agents believe that the central bank has a stable policy regime, so temporary inflation shocks do not cause large

revisions in beliefs about future inflation.

The standard macroeconomic model often assumes that agents have stable expectations about future inflation; however, recent empirical work shows that expectations are not always well anchored in practice. The literature suggests that a central reason is that expectations are sensitive to inflation shocks, forecast errors, and incoming inflation data. [Carvalho et al. \(2023\)](#) argue that expectations become unanchored when agents begin doubting the stability of the inflation regime, causing beliefs to be more sensitive to forecast errors and incoming inflation data. Like Carvalho, [Bems et al. \(2021\)](#) show that the degree of expectation anchoring affects inflation persistence. Their results suggest that when expectations are strongly anchored, inflation reacts less to shocks and returns more quickly to previous levels. This literature is important to this paper since it links weaker anchoring to larger and more persistent responses of inflation expectations after shocks. My paper contributes to this discussion by asking whether inflation expectations in a fixed exchange rate regime are less sensitive to inflation shocks than expectations in a more flexible inflation-targeting regime.

A related explanation for imperfect anchoring comes from the literature on selective attention and experience-based expectation formation. Households generally do not pay attention to all available macroeconomic information at all times; they usually pay more attention to inflation when it is high, personally salient, or tied to recent negative economic experiences. [Coibion and Gorodnichenko \(2025\)](#) shows that attention to monetary policy and inflation heavily fluctuates, with households becoming more attentive to inflation news when economic conditions deteriorate. [D’Acunto et al. \(2024\)](#) emphasizes that households are difficult for central banks to reach, and that consumers end up relying on partial information and personal price experiences to form inflation expectations. This is also consistent with the experience-based learning model, which shows that lived macroeconomic experiences can shape beliefs following an initial shock ([D’Acunto et al., 2021](#); [Malmendier and Nagel, 2011](#)). More recent work on selective memory also suggests that households may overweight recent

or more memorable inflation experiences while neglecting the more frequent experience of price stability ([Gennaioli et al., 2024](#)). The idea of selective attention is relevant to my paper because it provides a behavioral reason why inflation expectations move when central banks have formal policy frameworks designed to anchor them.

Building on this idea, my paper asks whether the monetary regime can reduce the unanchoring effects caused by selective attention. If households believe that the monetary regime is stable and credible, their trust that inflation will return to the target could cause them to feel less need to continuously update their beliefs in response to every inflation shock. In this case, greater credibility could make households less attentive to short-run macroeconomic fluctuations, causing less frequent updating of beliefs. There could also be an opposing explanation. A more credible regime could make monetary policy more salient and easier to understand, which would lead households to pay more attention to the central bank’s inflation objectives, respond more to policy, and respond less to news and noise. My paper therefore examines whether monetary regimes shape not only the level of inflation expectations, but whether it can influence the extent to which households selectively attend to inflation shocks.

Given the challenges of policymakers trying to anchor agents’ expectations, there has been literature that examines how central bank communication, trust, and economic knowledge shape inflation expectations. While previous models show that central banks can influence expectations through public communication and credibility building, recent research shows that households and firms often respond weakly to conventional monetary policy announcements, especially in low-inflation environments where inflation is not highly salient ([Coibion et al., 2020](#); [D’Acunto et al., 2024](#)). [D’Acunto et al. \(2024\)](#) emphasizes that households are difficult for central banks because they have limited knowledge of monetary policy and do not form expectations from complete information. Similarly, [Coibion et al. \(2020\)](#) argue that inflation expectations may serve as a useful policy tool, especially in the current era of

low interest rates. However, they also emphasize that this tool is not ready for deployment as the mechanisms linking expectations to behavior remain incompletely understood. This shows that current monetary policy frameworks alone are not enough to anchor expectations if households do not notice, understand, or trust the signals sent by the central bank. My paper asks whether a fixed exchange rate regime can act as an anchor that reduces the sensitivity of inflation expectations even when ordinary monetary policy communication faces limits.

With the idea of credibility in mind, a related strand of literature examines whether fixed exchange rates can act as external nominal anchors. According to the literature, a country is able to strengthen monetary credibility by tying domestic policy to a more credible external regime, thus making deviations from low inflation more costly and more visible. [Giavazzi and Pagano \(1988\)](#) argue that exchange rate commitments can help governments “tie their hands” by imposing discipline and making the public aware that unfounded inflationary policy would carry penalties in the form of exchange rate pressure and real appreciation. Beyond using a fixed exchange rate as a reputation channel, [Edwards \(1992\)](#) shows that credible exchange rate anchors can reduce inflationary inertia. This means that past inflation is less likely to persist and feed into future inflation once the commitment to the peg is viewed as credible. The ECB has also acted as a shock absorber by shaping clear expectations around its price stability objective ([Rostagno et al., 2019](#)). This literature is central to my paper because it frames fixed exchange rate regimes as a possible source of monetary credibility and expectation anchoring.

These theoretical arguments raise an empirical question: given that a fixed exchange rate regime provides an external nominal anchor, does this actually translate into a weaker change in inflation expectations following an inflation shock? To answer this question, my paper follows established empirical literature that uses impulse responses to study the dynamic effects of macroeconomic shocks. [Jordà \(2005\)](#) develops local projections as a flexible method

for estimating dynamic responses by horizon. This approach is useful for my paper because anchoring is a dynamic process, and local projections can help me answer how expectations respond to an inflation shock contemporaneously, but also whether the same response persists over subsequent quarters. This is useful for studying whether expectation responses persist after an inflation shock without having to create a full dynamic structure required by VARs. My identification strategy also follows the shock-based empirical literature. [Romer and Romer \(2004\)](#) shows the importance of isolating exogenous policy variation, and [Nakamura and Steinsson \(2018\)](#) similarly emphasizes the value of using identified surprises to estimate causal macroeconomic responses. In this paper, I use [Känzig \(2021\)](#) oil supply news shock as an external instrument in the first stage of an IV local projection framework, using exogenous oil-driven variation to estimate the response of inflation expectations over time.

3 Data

This paper examines the dynamics of inflation expectations of two highly comparable economies in Scandinavia: Denmark and Sweden. Both countries have similar levels of economic development, institutional styles, and data availability, which makes them a great candidate for empirical comparisons. The key difference lies in both country's monetary framework. Denmark uses a fixed exchange rate, pegging its currency to the euro, while Sweden operates under a floating exchange rate with an independent monetary policy. This difference gives way to a natural setting to evaluate how inflation expectations respond to shocks under alternative monetary systems. To achieve this, this paper will use data on inflation, inflation expectations, and relevant controls to compare the difference in inflation expectations across the two regimes.

3.1 Data Sources

This paper uses credible macroeconomic data for Denmark and Sweden drawn from official government sources, statistical agencies, and international databases. The dataset includes measures of inflation, inflation expectations, monetary policy, labor market conditions, and global shocks. To ensure comparability, all are aligned to quarterly frequency.

3.1.1 Inflation

Inflation is measured using the seasonally adjusted change in the consumer price index. CPI data for both Denmark and Sweden have been obtained through OECD, the Organisation for Economic Co-operation and Development. For Sweden, I had the choice between using CPI and CPIF, which excludes changes in mortgage interest rates. While the Swedish government uses CPIF for policy, I chose CPI for two reasons: (1) to maintain consistency with the Danish inflation series, and (2) because CPI better captures the broad inflation experience faced by households. This makes it more appropriate for an analysis focused on actual consumer inflation outcomes relative to inflation expectations.

3.1.2 Inflation Expectations

Inflation expectations is a one-year forward-looking measure, which asks consumers how they expect prices to develop over the next 12 months relative to the previous 12 months. To standardize, both Denmark and Sweden's expectations data comes from the European Commission's Business and Consumer surveys. More specifically I use question 6 under the Consumers Qualitative and Quantitative Inflation perceptions section. The survey asks whether consumers expect prices to: 1) increase more rapidly (PP), 2) increase at the same rate (P), 3) increase at a slower rate (E), 4) stay about the same (M), or 5) fall (MM). The data is reported as a balance statistic, calculated from the distribution of survey responses

as:

$$\text{Balance} = PP + \frac{P}{2} - \frac{M}{2} - MM$$

The balance statistic ranges from -100 to 100 , where higher values indicate that more consumers expect prices to rise faster over the next year. The survey is collected monthly, so to get a quarterly statistic, I take an average across each quarter.

A limitation of this measure is that the balance statistic does not report consumers' actual predicted inflation rate. Given that the results are compressed into broad qualitative categories, respondents do not report a numerical inflation forecast; the measure cannot tell how large consumers expect inflation to be. This means that a higher value should be interpreted as indicating that more consumers expect prices to rise faster, not that consumers expect inflation to be a specific percentage point higher. As a result, the measure is useful for comparing whether inflation expectations in Denmark is relatively more or less sensitive to inflation shocks compared to Sweden, but it is less precise than a numerical measure of expected inflation.

One additional consideration is that the survey measures expectations over the next 12 months, while some studies of anchoring focus on longer-horizon expectations. Since longer-term expectations are often more closely linked to beliefs about the credibility of the monetary regime, the one-year-ahead measure should be interpreted as capturing the sensitivity of near-term expectations to inflation shocks rather than a complete measure of long-run anchoring.

3.1.3 Oil Shock

I use oil supply news shocks as an instrument for realized inflation because inflation is endogenous to the same macroeconomic conditions that shape inflation expectations. For example, both inflation and inflation expectations can be impacted by domestic demand,

monetary policy, or labor market conditions. If I use realized inflation as the shock term, the estimated relationship may capture the effect of inflation and the effect of the omitted macroeconomic variables.

The purpose of [Känzig \(2021\)](#) oil supply news shock is to isolate inflation variation. Change in oil prices is not ideal since oil prices could be contaminated by oil supply and global demand changes. [Känzig \(2021\)](#) shock is constructed to capture revisions in expectations in oil supply through surprise shocks, which is a cleaner source of external variation than change in oil prices.

This shock should be exogenous; while Denmark and Sweden are well-developed economies, they are small enough that they are unlikely to influence global oil supply news or OPEC-related expectations. Given the oil supply news shock is constructed using global oil markets instead of Danish or Swedish-specific data, it provides external variation in realized inflation. Thus I use it to isolate the component of inflation driven by global supply shocks.

3.1.4 Policy Rate

The monetary policy stance is proxied by the central bank policy rate. For Denmark, the data comes from Danmarks Nationalbank, the Danish central bank, where I use the official lending rate series. The data for the Swedish policy rate comes from Sveriges Riksbank, the central bank of Sweden. The raw dataset is updated only when the policy rate changes, not at a quarterly rate. I therefore assign the applicable policy rate to each date and then aggregate the series to the quarterly level. These rates reflect the primary instrument used by each country's central bank to influence short-term interest rates.

3.1.5 Unemployment

Labor market conditions are captured using the unemployment rate. Both Denmark and Sweden's unemployment rates come from OECD, The Organisation for Economic Co-

operation and Development. The dataset is monthly, so to get a quarterly statistic, I take an average across each quarter. This dataset provides a control for labor market conditions in both countries.

Table 1: Data Sources and Variable Construction

Variable	Denmark Source	Sweden Source	Original Frequency	Transformation	Final Measure
CPI	OECD	OECD	Monthly	Quarterly average & seasonal adjustment	Seasonally Adjusted Inflation Rate
Inflation Expectations	EC Consumer Survey	EC Consumer Survey	Monthly	Quarterly average	Balance statistic
Känzig Oil Shock	X	X	Quarterly	Aligned to quarter	IV for inflation
Policy Rate	Danmarks Nationalbank	Sveriges Riksbank	Monthly	Quarterly average	Policy rate
Unemployment	OECD	OECD	Monthly	Quarterly average	Unemployment rate

3.2 Descriptive Statistics and Context

This study looks at data from 1995Q4 through 2025Q2. I start at 1995Q4 because that is as far as the Sweden Inflation Expectation series goes. Prior to 1999Q1, the Danish Krone was under the European Exchange Rate Mechanism (ERM), which was effectively pegged to the Deutsche Mark. There was a consideration to start the sample in 2002Q1 since the Euro was introduced on January 1, 1999, and started physical circulation on January 1, 2002, and excluding the earlier period could avoid introducing a regime transition which would affect inflation expectations for reasons unrelated to the exchange rate arrangement itself. However, I decided that the impact from this transition would be minimal and that the extra observations would be beneficial to the accuracy of the study.

Figure 1 plots the time-series behavior of the main variables used in the analysis. Panels A, C, and D show that Denmark and Sweden share broadly similar macroeconomic dynamics

Time-Series Dynamics of Key Variables, 1995Q4-2025Q2



Figure 1: Time-series dynamics of key macroeconomic variables, 1995Q4-2025Q2. Panels A through D show inflation, inflation expectations, policy rates, and unemployment rates for Denmark and Sweden, while Panel E shows the Känzig oil supply news shock used as the instrument for realized inflation. Inflation is measured as seasonally adjusted quarter-on-quarter annualized CPI inflation, and inflation expectations are measured as a European Commission consumer survey balance statistic.

over the sample period. Inflation follows a comparable path in both countries, which is quite moderate for much of the sample before spiking following the 2021 inflation surge. Both countries have policy rates that move closely together, with both countries beginning the sample with relatively high rates, experiencing a long decline through the 2000s and 2010s, and then raising rates sharply after 2021. Sweden generally exhibits a higher unemployment rate than Denmark, but both series move in the same general direction during major macroeconomic episodes.

Panel B shows a more pronounced difference in inflation expectations. Sweden’s inflation expectations are generally higher than Denmark’s for much of the sample, particularly in the earlier years, while Denmark’s series remains lower on average. While both Sweden and Denmark see downward spikes in inflation expectations, Sweden’s balance statistic stays positive, whereas Denmark’s falls to substantially negative values. Over time, however, the two expectation series become more similar, and both respond strongly around the post-pandemic inflation episode. Finally, Panel E plots the Känzig oil supply news shock, which is common to both countries and provides the external variation used to instrument realized inflation in the IV local projection framework. This is especially relevant during periods in late 2014 and the early 2020s, when large oil price movements contributed to inflationary pressures and made global supply shocks important for inflation dynamics. Overall, the figure suggests that Denmark and Sweden are exposed to many of the same broad macroeconomic forces, while still differing in the level and behavior of inflation expectations, which is central to the empirical question of this paper.

In the post 1995 sample, both countries have comparable average seasonally adjusted quarterly inflation rates, with Danish inflation averaging around 1.92% and Swedish inflation at 1.65%. However, inflation is more volatile in Sweden, with a standard deviation of 2.89 compared to a standard deviation of just 2.27 in Denmark. Both countries see inflation rise sharply during the post-pandemic/Russia-Ukraine inflation surge, with Denmark reaching

Table 2: Summary Statistics, 1995Q4–2025Q2

Variable	Denmark				Sweden			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Inflation	1.92	2.27	-2.16	12.99	1.65	2.89	-5.76	12.90
Inflation expectations	6.13	17.27	-37.07	41.95	21.90	13.32	-13.48	51.09
Policy rate	2.01	1.74	-0.45	5.40	2.06	1.97	-0.50	8.91
Unemployment	5.57	1.19	3.43	8.23	7.57	1.10	5.10	10.30
Oil shock (same)	-0.01	1.09	-4.43	3.23	-0.01	1.09	-4.43	3.23

12.99% and Sweden reaching 12.90% in 2022Q2.

The inflation expectations data sees much more variation. The average inflation expectations are 6.13 in Denmark and 21.90 in Sweden, which indicates that, on average, a larger share of Swedish respondents are expecting prices to rise more quickly over the next year. The control variables also show similarities and differences across the two economies. The policy rates are similar on average, at 2.01% in Denmark and 2.06% in Sweden, while unemployment is lower on average in Denmark 5.57% versus Sweden 7.57%.

Altogether, the data support the use of Denmark and Sweden as a comparative setting: the two countries have similar average inflation and policy-rate environments, while differing in exchange rate regime, inflation volatility, and the level of inflation expectations. The empirical analysis will use impulse responses to examine whether these differences are reflected in how inflation expectations respond to inflation shocks.

4 Model and Methodology

This paper uses an Instrumental Variable Local Projections (IV-LP) model to estimate the dynamic response of inflation expectations to inflation shocks in Denmark and Sweden. The empirical reason for looking at Denmark and Sweden is to compare how inflation expectations respond under two different monetary regimes: Denmark’s fixed exchange rate

regime versus Sweden’s floating exchange rate regime. Instead of estimating the entire multivariate system and deriving impulse responses indirectly through VARs, this paper estimates inflation expectations at each forecast horizon. This section begins by outlining the empirical specification, then presents the estimation and inference procedures, and concludes by explaining how the results are interpreted.

4.1 Empirical Specification: Instrumental Variable Local Projections

This section describes the empirical specification used to estimate the dynamic response of inflation expectations to inflation shocks. The procedure is estimated separately for each country using an augmented local projection framework, which incorporates an instrumental variable to address endogeneity concerns.

Local projections ([Jordà, 2005](#)) offer a straightforward method to estimate impulse responses for each horizon. The main difference from a full dynamic system, such as VAR, is that a new regression is estimated for each forecast horizon. This is convenient because an impulse response function forecasts at increasingly distant horizons, which means that a mis-specification in the global model can compound when the horizon increases. By conducting regressions that are local to each individual horizon, the design of the estimation is able to match each variable more closely. Local projections are straightforward to estimate using standard regression methods and allow for direct analysis of the coefficients of interest.

The main challenge in estimating the effect of inflation on inflation expectations is that realized inflation may be endogenous. Inflation could be correlated with unobserved shocks, which could directly impact expectations. To address this issue I use an instrumental variable local projection (IV-LP) approach. This method uses an external instrument to isolate exogenous variation in inflation, assuming that the instrument affects inflation expectations

only through realized inflation. Using instrumented inflation, we will be able to analyze its effect on inflation expectations at each horizon.

For each country, the specification estimates future inflation expectations by regressing them on instrumented current inflation, lagged inflation, lagged inflation expectations, and a set of macroeconomic controls. Inflation at time t is instrumented using oil shocks to isolate exogenous variation.

4.1.1 First Stage

In the first stage, realized inflation is regressed on the instrumental variable and a set of controls:

$$\pi_t = \gamma Z_t + \sum_{s=1}^N \phi_s \pi_{t-s} + \sum_{k=1}^M \psi_k Y_{t-k} + \delta R_t + \eta U_t + \rho T_t + v_t \quad (3)$$

where π_t denotes realized inflation at time t , and Z_t is the instrumental variable used to isolate exogenous variation in inflation, in this case representing oil shocks taken from [Känzig \(2021\)](#). The terms π_{t-s} represent lagged inflation for periods $t-1$ to $t-N$. Y_{t-k} represents the lagged inflation expectations for periods $t-1$ to $t-M$. A reasonable choice for N and M is 4, so specifications include one year of quarterly lags for inflation and inflation expectations. An important note: Only contemporaneous inflation is treated as endogenous and instrumented; lagged inflation and lagged expectations are included as predetermined controls.

The remaining variables are macroeconomic controls intended to isolate exogenous variation in inflation. R_t is the policy rate and U_t is the unemployment rate. T_t denotes a linear time trend, included to capture slow-moving structural changes in the economy. v_t is the error term.

4.1.2 Second Stage

Let $h = 0, 1, \dots, H$ denote the number of quarters following each observation.

$$Y_{t+h} = \alpha_0^{(h)} \hat{\pi}_t + \sum_{s=1}^N \alpha_s^{(h)} \pi_{t-s} + \sum_{k=1}^M \beta_k^{(h)} Y_{t-k} + \delta^{(h)} R_t + \eta^{(h)} U_t + \rho^{(h)} T_t + \epsilon_{t+h} \quad (4)$$

where Y_{t+h} denotes inflation expectations at horizon h , and $\hat{\pi}_t$ is the fitted value of inflation from the first stage. The coefficient $\alpha_0^{(h)}$ traces out the impulse response of inflation expectations following an exogenous inflation shock at horizon h . The remaining terms are as defined in Equation 3, ϵ_{t+h} is the error term.

4.2 Estimation, Inference and Cross-Country Comparison

The IV-LP model yields a sequence of horizon-specific coefficients on which together trace out the impulse response of inflation expectations to an inflation shock. The main coefficient of interest is $\alpha_0^{(h)}$, which measures the response of inflation expectations at horizon h to contemporaneous inflation at time t . The sequence $\alpha_0^{(0)}, \alpha_0^{(1)}, \dots, \alpha_0^{(H)}$ thus tracks the dynamic response of inflation expectations to an inflation shock.

The paper uses HAC standard errors to construct 95% confidence intervals around each estimated impulse response coefficient. This allows statistical significance to be verified at each horizon, and a measure of the precision of each response. The end product will be a set of impulse response functions, represented as coefficients, that summarize how inflation expectations evolve in the quarters following an inflation shock. For each country this regression is run, the responses will be plotted with their associated confidence bands to show the magnitude of the response and the degree of statistical uncertainty.

I run the same IV-LP framework separately for Denmark and Sweden. Denmark serves as the fixed exchange rate case, while Sweden serves as the floating exchange rate case. I then compare the estimated impulse response functions across countries. If Denmark's

response is smaller or less persistent than Sweden's, this would be consistent with stronger anchoring under the fixed exchange rate regime. If the response paths are similar, then the results would not support the hypothesis that Denmark's fixed exchange rate regime produces stronger anchoring.

4.3 Interpretation of the Impulse Responses

The impulse response functions give us a clear picture of the dynamic effect that an inflation shock has on inflation expectations. In the figures, the horizontal axis represents the number of quarters after the shock, given by $h = 0, 1, \dots, H$. The vertical axis measures the estimated response of inflation expectations. A response that is close to zero implies that expectations are relatively well anchored, while a deviation from zero with higher persistence or of larger magnitude suggests that expectations adjust more strongly following a shock.

The impulse responses are interpreted by examining both the size and persistence of the estimated response. If inflation expectations are well anchored, the response to an inflation shock should be small and should return toward zero relatively quickly. Larger or more persistent responses suggest that expectations adjust more strongly to realized inflation. Statistical significance is assessed using 95% confidence intervals; if the confidence interval excludes zero at a given horizon, the response is statistically significant at that horizon.

This paper also tests whether the response paths differ significantly from one another over the full impulse response horizon. To accomplish this, I conduct a joint Wald test of whether the impulse response is equal to zero over the set horizon path. This avoids relying only on visual comparisons or separate point-wise tests at each horizon.

For the Wald test, I first construct the difference in impulse responses across countries at each horizon. Specifically, the coefficients on the interaction terms capture the difference

between Denmark and Sweden:

$$\alpha_{0,\text{diff}}^{(h)} = \alpha_{0,\text{DEN}}^{(h)} - \alpha_{0,\text{SWE}}^{(h)}, \quad h = 0, 1, \dots, H. \quad (5)$$

These coefficients, therefore, trace the difference between the two impulse response functions at each horizon. Formally, the null hypothesis is that the cross-country differences in the impulse response coefficients are jointly equal to zero over the selected horizons:

$$H_0 : \alpha_{0,\text{diff}}^{(0)} = \alpha_{0,\text{diff}}^{(1)} = \dots = \alpha_{0,\text{diff}}^{(H)} = 0. \quad (6)$$

Equivalently, this can be written in matrix form as:

$$H_0 : R\hat{\alpha}_{0,\text{diff}} = 0, \quad (7)$$

where R is the restriction matrix that selects the impulse response coefficients being tested.

The corresponding Wald statistic is:

$$W = (R\hat{\alpha}_{0,\text{diff}})' \left[R\widehat{\text{Var}}(\hat{\alpha}_{0,\text{diff}})R' \right]^{-1} (R\hat{\alpha}_{0,\text{diff}}). \quad (8)$$

Under the null hypothesis, the Wald statistic is asymptotically distributed as:

$$W \sim \chi^2(q), \quad (9)$$

where q is the number of restrictions tested.

Since $H = 16$, the impulse response is estimated over 17 quarters ($h = 0, \dots, 16$). The joint Wald test, therefore, imposes restrictions across all horizons:

$$H_0 : \alpha_{0,\text{diff}}^{(0)} = \alpha_{0,\text{diff}}^{(1)} = \dots = \alpha_{0,\text{diff}}^{(16)} = 0. \quad (10)$$

This corresponds to $q = 17$ restrictions, so that the Wald statistic is given by:

$$W \sim \chi^2(17). \quad (11)$$

Overall, these impulse responses are the core results needed to answer the paper’s empirical question. If the regression run for Denmark exhibits smaller and less persistent responses, this would be in line with the view that inflation expectations are more firmly anchored under a fixed exchange rate regime. However, if the responses are similar across both countries, or if Sweden’s responses are smaller or less persistent, the hypothesis would fail.

5 Results

5.1 Main IV-LP Results

This section analyzes the set of impulse response functions of inflation expectations to an inflation shock using the Instrumental Variable Local Projection framework. The baseline model uses $H = 16$, $N = 4$, and $M = 4$ in Equations (3) and (4). $N = 4$ includes $s = 0, \dots, 4$, the inflation terms cover five quarters in total: current inflation and four lagged quarters. $M = 4$ includes $k = 1, \dots, 4$, the lagged expectations terms cover the previous four quarters.

I first examine whether the oil supply news shock is a relevant instrument for realized inflation. Table 3 reports the first-stage results for Denmark and Sweden. The coefficient on the oil shock indicates how strongly global oil supply news shocks predict domestic inflation after controlling for lagged inflation, lagged inflation expectations, and macroeconomic controls.

Table 3: First-Stage IV Results 1995Q4-2025Q2

	Denmark inflation	Sweden inflation
Oil supply news shock	0.436***	0.379**
Standard error	(0.135)	(0.161)
<i>p</i> -value	0.002	0.021
Lagged inflation controls	Yes	Yes
Lagged expectations controls	Yes	Yes
Policy rate control	Yes	Yes
Unemployment control	Yes	Yes
Time trend	Yes	Yes
Observations	115	115
First-stage F-statistic	10.46	5.51
R^2	0.456	0.511

Notes: HAC robust standard errors are shown in parentheses. The *p*-value refers to the coefficient on the Känzig oil supply news shock. The first-stage F-statistic refers to the excluded instrument test for the Känzig oil supply news shock. Both specifications include four lags of inflation, four lags of inflation expectations, the policy rate, the unemployment rate, and a linear time trend. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 3 reports the first-stage IV results, showing whether the Känzig oil supply news shock is a relevant instrument for realized inflation. The coefficient on the oil shock is positive and statistically significant for both countries, which implies that positive oil supply news shocks are associated with higher measured inflation in the first stage. For Denmark, the coefficient is 0.436 with a *p*-value of 0.002, while for Sweden it is 0.379 with a *p*-value of 0.021. This suggests that the instrument has predictive power for inflation in both cases. However, the first-stage F-statistic is stronger for Denmark at 10.46, while Sweden's F-statistic is lower at 5.51. The lower Swedish first-stage F-statistic indicates that the instrument is weaker for Sweden, which is a limitation when interpreting the Swedish IV-LP estimates. This may partly explain the wider confidence intervals for Sweden in the impulse response estimates. The R^2 values of 0.456 for Denmark and 0.511 for Sweden suggest that the full first-stage specifications explain a meaningful share of inflation variation.

Table 4: Baseline IV-LP Results by Horizon, 1995Q4–2025Q2

Horizon	Denmark			Sweden		
	Coefficient	SE	<i>p</i> -value	Coefficient	SE	<i>p</i> -value
0	2.969***	(1.124)	0.008	6.070**	(2.770)	0.028
1	4.911***	(1.781)	0.006	8.780**	(3.650)	0.016
2	0.653	(1.605)	0.684	2.530	(3.177)	0.426
3	-1.679	(2.230)	0.452	0.474	(3.807)	0.901
4	-1.636	(2.584)	0.526	-1.963	(3.829)	0.608
5	-5.728*	(3.114)	0.066	-4.559	(4.122)	0.269
6	-5.035*	(2.856)	0.078	-3.223	(3.984)	0.419
7	-4.042	(2.481)	0.103	-2.935	(3.984)	0.461
8	-1.846	(2.210)	0.404	-4.060	(4.168)	0.330
9	-2.396	(2.351)	0.308	-3.484	(3.686)	0.344
10	1.849	(3.075)	0.547	-2.135	(3.699)	0.564
11	4.967*	(2.919)	0.089	0.943	(3.780)	0.803
12	5.074	(4.133)	0.220	1.384	(3.544)	0.696
13	4.882	(3.071)	0.112	2.084	(3.822)	0.586
14	3.336	(2.529)	0.187	-2.365	(2.969)	0.426
15	2.425	(2.212)	0.273	-3.659	(3.959)	0.355
16	-0.294	(2.330)	0.900	-2.875	(3.857)	0.456

Notes: The table reports IV-LP estimates by horizon for the 1995Q4–2025Q2 sample. HAC robust standard errors are shown in parentheses. The *p*-values correspond to the coefficient estimate for each country at each horizon. Both specifications use the Känzig oil supply news shock as an instrument for inflation and include the baseline controls. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. It should be noted that the reported *p*-values test whether each country's estimated response is statistically different from zero, not whether Denmark's and Sweden's inflation expectation responses are statistically different from each other.

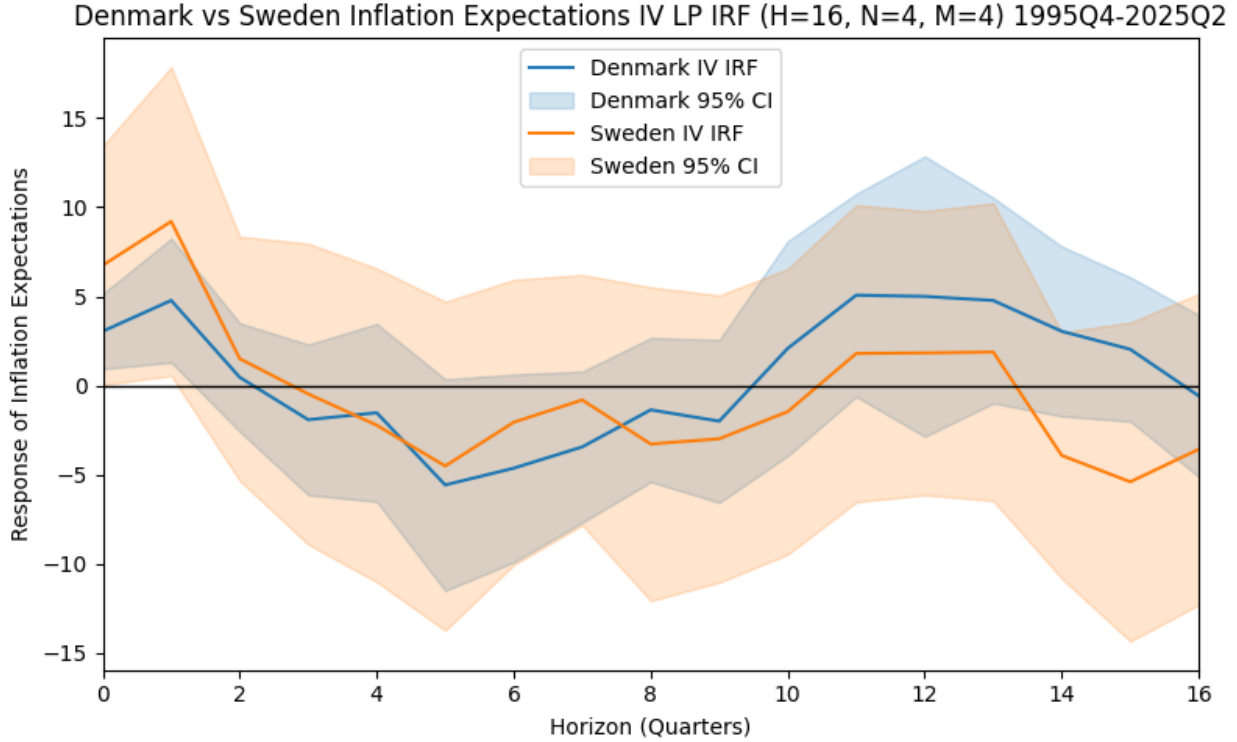


Figure 2: This figure plots local projection impulse responses of inflation expectations to a one-unit inflation shock for Denmark and Sweden over quarters $h = 0, \dots, 16$ after an inflation shock using data from 1995Q4 through 2025Q2. The solid line represents point estimates while the shaded regions are 95% confidence intervals. The x-axis shows the number of quarters after the shock, and the vertical axis shows the response of inflation expectations. This specification uses 4 quarter lags of inflation and 4 quarter lags of inflation expectations, and controls for policy rate, unemployment rate, and trend.

Figure 2 plots the estimated impulse responses for Denmark and Sweden along with 95% confidence intervals. Looking at Table 4, at horizon 0, both countries exhibit a positive point estimate for the response of inflation expectations to the inflation shock. Sweden's initial response is larger, reaching a higher positive value at $h = 1$, while Denmark's response is more moderate right after the shock. However, the confidence intervals are relatively wide, especially for Sweden, meaning that the initial difference between the two countries should be interpreted cautiously.

Both Denmark and Sweden's responses decline quite quickly, with Denmark's inflation expectation impulse response turning negative $h = 3$, before Sweden's impulse response

$h = 4$. The medium horizons $h = 4$ to $h = 9$ seem quite similar for both Denmark and Sweden, where the negative IRFs float between -1.5 and -6 . At longer horizons, Denmark's response turns positive at $h = 10$, one period earlier than Sweden's response at $h = 11$. Denmark also reaches a higher peak between $h = 11$ and $h = 13$, before both IRFs appear to converge back toward zero.

The large confidence intervals likely reflect the relatively low ratio of usable observations to the number of regression variables. Although the dataset has three decades' worth of data, the effective regression sample becomes much smaller after restricting the sample period, including four lags of inflation and inflation expectations, adding macroeconomic controls, and estimating responses at longer horizons. In addition, an IV regression generally produces wider confidence intervals than OLS because it relies only on the component of inflation explained by the instrument, rather than all variation in inflation. This issue is amplified in the local projections framework since a new regression is run for each horizon. While the wide confidence intervals are not necessarily evidence that the point estimates are wrong, they do mean that the estimates are subject to a degree of uncertainty.

These results suggest that Sweden has a stronger immediate point-estimate response to inflation shocks, while Denmark's response becomes more positive and persistent at longer horizons. However, the confidence intervals substantially overlap across most horizons, and many horizon-specific confidence intervals include zero. Thus these differences should not be interpreted as strong statistical evidence that inflation expectations are differently anchored across the two regimes.

5.2 Differences Across Regimes, Significance Testing

To formally assess whether responses differ across Denmark and Sweden, I compare the impulse response functions presented in Figure 2 and conduct a joint test of equality of the

response paths across all horizons. As shown in the previous section, Sweden exhibits a larger positive initial response at short horizons ($h = 0, 1$), while Denmark's response becomes more persistent and increases in magnitude at medium and longer horizons. In particular, Denmark's response turns positive again after the mid-horizon decline and remains above Sweden's response from approximately $h = 10$ to $h = 13$. In contrast, Sweden's response declines more quickly and remains closer to zero across much of the horizon.

To evaluate whether the response paths are statistically significant, I conduct a joint test of the null hypothesis that the impulse response paths are equal across Denmark and Sweden at all horizons. This test is implemented using a pooled local projection specification, more specifically, a Wald test, which evaluates the entire response path jointly as in Equations 10 and 11.

Table 5: Full-Path Equality-of-Paths Wald Test, 1995Q4–2025Q2

Horizon range	Horizons	Wald statistic	df	p-value
$h = 0$ to $h = 16$	17	9.476	17	0.924

Notes: This table reports a Wald test of whether Denmark and Sweden IV-LP response paths are jointly equal across all horizons from $h = 0$ to $h = 16$. The column “df” reports the degrees of freedom, which equals the number of joint restrictions tested. Since the p-value is high, I fail to reject the null hypothesis that the two response paths are equal.

The Wald test yields a p-value of 0.924, which means I do not reject the null hypothesis. This means the results do not provide statistical evidence that the impulse response paths differ across regimes.

5.3 Robustness Check: Alternative Data Sample

As a robustness check for my initial results, I re-estimate the model but I exclude data following 2019Q4 so that the abnormal impacts of COVID do not distort the estimated impulse responses.

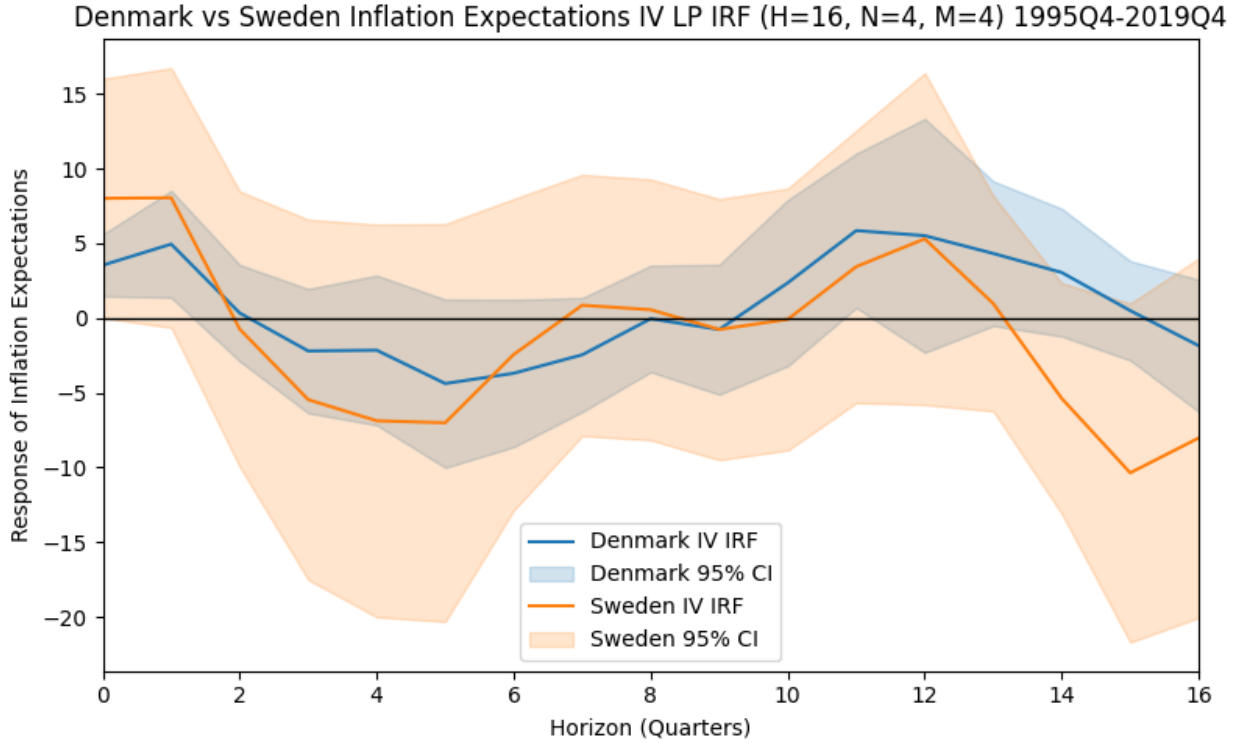


Figure 3: This figure plots local projection impulse responses of inflation expectations to a one-unit inflation shock for Denmark and Sweden over quarters $h = 0, \dots, 16$ after an inflation shock using data from 1995Q4 through 2019Q4. The solid line represents point estimates while the shaded regions are 95% confidence intervals. The x-axis shows the number of quarters after the shock, and the vertical axis is the response of inflation expectations. This specification uses 4 quarter lags of inflation and 4 quarter lags of inflation expectations, and controls for policy rate, unemployment rate, and trend.

The impulse responses from my robustness check are quite similar to those obtained in my initial test. As in the baseline results, Sweden exhibits a larger initial response at short horizons, while Denmark's response becomes more persistent and increases in magnitude at medium horizons. In both samples, Denmark displays a stronger positive response at medium horizons, peaking around $h = 10$ to $h = 12$, whereas Sweden's response remains

closer to zero and is less persistent. Both impulse response functions seem to converge towards the direction 0 by $h = 16$.

The main difference is the size of the confidence intervals. Relative to the baseline results, Denmark’s confidence intervals appear narrower in the medium to longer horizons, especially around $h = 8$ to $h = 13$, suggesting more precision for Danish results in the pre-2019 sample. Sweden’s confidence intervals, however, remain quite wide. This is quite apparent right after the inflation shock at $h = 0$ to $h = 2$, the medium horizons around $h = 4$ to $h = 5$, and the later horizons from roughly $h = 14$ to $h = 16$. As a result, Sweden’s impulse responses should be interpreted with caution, since the larger uncertainty bands make it difficult to distinguish impulse responses from Denmark.

Table 6: Full-Path Equality-of-Paths Wald Test, 1995Q4–2019Q4

Horizon range	Horizons	Wald statistic	df	p-value
$h = 0$ to $h = 16$	17	11.558607	17	0.826

Notes: This table reports a Wald test of whether the Denmark and Sweden IV-LP response paths are jointly equal across all horizons from $h = 0$ to $h = 16$. The column “df” reports the degrees of freedom, which equals the number of joint restrictions tested. Since the p-value is high, I fail to reject the null hypothesis that the two response paths are equal.

To evaluate if response paths differ across regimes in this restricted sample, I conduct the same Wald test. The resulting p-value is 0.826, so I again fail to reject the null hypothesis of equal response paths. The test continues to suggest that the impulse response paths are not statistically distinguishable across Denmark and Sweden.

Overall, the comparison suggests that the inclusion of the post-2019 period does not materially alter the main conclusions.

5.4 Summary of Results

The results provide evidence that inflation expectations in Denmark and Sweden respond quite similarly to inflation shocks. While there are some differences in timing and magnitude, the overall impulse response paths do not differ in a statistically significant way. The joint Wald test fails to reject the null hypothesis, indicating that the differences are not statistically meaningful.

My results do not provide evidence that the fixed exchange rate regime produces more strongly anchored inflation expectations than the floating exchange rate regime in this Denmark-Sweden comparison.

One possible interpretation is that Denmark's peg to the euro does not fully insulate inflation expectations from global shocks. Because many of the major inflation shocks in the sample are global, the euro is impacted as well, hence both Denmark and Sweden may be exposed to similar inflationary pressures regardless of the exchange rate regime. A fixed exchange rate could still have a different impact on inflation expectations for local shocks, but the results here suggest that it does not generate significantly different expectation dynamics in response to the global shocks observed in this sample.

Another interpretation is that Sweden's floating exchange rate regime is supported by sufficiently strong monetary institutions and high degree of policy credibility. If both countries face similar shocks and both central banks respond in ways that households view as credible, expectations could have similar levels of anchoring despite different exchange rate regimes. Further research could be done to see if differences between fixed and floating regimes might be larger in countries where the floating-regime central bank has weaker credibility or less institutional trust.

Overall, the evidence suggests that inflation expectations behave similarly across the two regimes, and that the presence of a fixed exchange rate alone does not guarantee more

strongly anchored expectations.

6 Conclusion

This paper asks whether inflation expectations are better anchored under fixed exchange rate regimes by comparing Denmark’s fixed exchange rate regime with Sweden’s floating exchange rate regime. The theoretical motivation is that a fixed exchange rate could serve as an external nominal anchor by tying domestic monetary policy to a more credible monetary authority, making households less likely to alter expectations in response to inflation shocks. To test this, I use an instrumental variable local projection framework in which inflation expectations are regressed on realized inflation, with realized inflation instrumented using oil supply news shocks. I then compare the resulting impulse response paths across Denmark and Sweden and test whether the two response paths differ across countries. The results do not provide evidence that inflation expectations are more strongly anchored under Denmark’s fixed exchange rate regime than under Sweden’s floating regime.

These findings suggest that a fixed exchange rate alone does not necessarily guarantee stronger anchoring of inflation expectations. One reason may be the type of shock used in this paper. The instrument captures global oil supply news shocks rather than country-specific disturbances. Because these shocks affect Denmark, Sweden, and the euro area at the same time, Denmark’s peg to the euro may not provide a clear insulation mechanism in this setting. As a result, Denmark’s fixed exchange rate regime may not generate a meaningfully different expectation response from Sweden’s floating regime. However, these results speak most directly to global inflation shocks and leave open whether fixed exchange rates provide stronger anchoring in response to local or domestic shocks.

Another interpretation is that Sweden’s floating exchange rate is supported by credible monetary institutions. While Sweden doesn’t use a fixed exchange rate, households may

still trust its inflation-targeting central bank enough for expectations to remain similarly anchored relative to Denmark. In addition, Denmark and Sweden also have similar policy-rate environments, which may make their monetary regimes look more similar in practice. This implies that the credibility of the monetary institution may matter as much as the exchange rate regime itself.

Regarding policy implications, the results suggest that countries with credible monetary institutions may gain limited additional benefits from adopting a fixed exchange rate solely for the purpose of anchoring inflation expectations. In the Denmark-Sweden case study, Sweden's floating exchange rate regime appears to anchor expectations just as well as Denmark's fixed exchange rate regime. This suggests that institutional credibility could substitute for an external nominal anchor. However, countries may still choose to fix their exchange rate for other reasons, such as trade stability, financial integration, or reducing exchange rate volatility.

Overall, the evidence suggests that inflation expectations behave quite similarly across Denmark and Sweden after inflation shocks, given their different exchange rate regimes. Monetary credibility appears to depend not only on regime design but also on the type of shock and the credibility of the domestic institutions. One limitation is that the inflation expectations measure is based on a balance statistic, so the results should be interpreted as directional changes instead of percentage-point changes in expected inflation. Future research could extend this analysis by examining a more diverse set of countries, using numerical measures of inflation expectations, or comparing how local and global shocks affect inflation expectations under fixed versus floating exchange rate regimes.

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